

Capstone Project-MediAssist AI

Enterprise Healthcare Agentic AI Copilot

1. Problem Statement

Healthcare organizations store patient records, documents, reports, and operational data across multiple disconnected systems, making information retrieval slow and inefficient. There is a need for an AI-powered healthcare copilot that can unify structured, unstructured, and multimodal data to provide accurate, contextual, and intelligent assistance for healthcare workflows.

2. Stakeholders

- Doctors
- Hospital Administrators
- Front Desk Staff
- Insurance Team
- Clinical Operations Team
- Patients

3. Main Pain Points

- Fragmented healthcare data across multiple systems.
- Slow access to patient records, reports, and operational information.
- Manual appointment scheduling and repetitive administrative tasks.
- Inefficient billing and insurance claim verification processes.
- Difficulty in monitoring workflows and managing escalations.
- Poor interoperability between structured and unstructured data sources.
- Limited support for analyzing medical reports and prescription images.

4. Key Features

- Conversational Enterprise Search (RAG)
- Multi-Agent AI Architecture
- Multimodal AI (Image and Report Understanding)
- PostgreSQL Integration via MCP
- Workflow Automation
- Context-Aware Recommendations
- Analytical Insights and Reporting
- LLM-as-a-Judge Evaluation
- Dockerized Deployment
- Azure Cloud Hosting

5. Data Sources

A. Unstructured Data:

Hospital SOPs, Clinical Workflow Documents, Insurance Policies, Healthcare Compliance Guidelines, Operational Manuals, Discharge Summaries, and Patient Care Guidelines.

B. Multimodal Data:

Prescription Images, Medical Reports, and Laboratory Report Images.

C. Structured Data:

Patient Records, Admissions, Appointments, Doctor Information, Department Details, Billing Records, Laboratory Results, Prescription Records, Insurance Information, and Hospital Branch Data.

D. Operational Knowledge:

Clinical Workflows, Standard Operating Procedures, Department-Level Processes, and Healthcare Operational Guidelines.

6. Sample User Queries

Operational Queries

- Show patient history .
- Retrieve upcoming appointments.
- Get billing information for a patient.
- Search doctor schedules.

Analytical Queries

- Compare appointment volumes across hospital branches.
- Analyze insurance claims by provider.
- Show patients with abnormal HbA1c results.
- Identify prescription patterns across specialties.

Multi-Agent Queries

- Provide a complete patient summary and upcoming appointments.
- Generate follow-up recommendations based on reports and medical history.

Multimodal Queries

- Extract medicines from this prescription image.
- Summarize the uploaded medical report.
- Analyze the scanned prescription and provide recommendations.